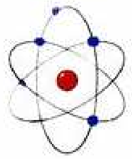
1.15 CHAPTER 1 SUMMARY – FROM ATOM TO MR IMAGE

Putting it all together: How MRI signal is created, evolves, and becomes an image.

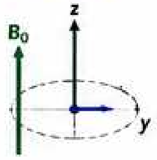
THE MRI SIGNAL FORMATION JOURNEY – STEP BY STEP

1. HYDROGEN
The Source



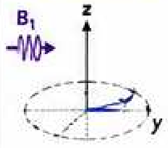
Hydrogen has one proton (spin 1/2). Protons behave like tiny magnets.

2. MAGNETIZATION
In B_0 Field



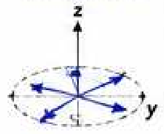
In a static magnetic field (B_0), spins align with (low energy) or against (high energy) B_0. Net magnetization M_0 is along +z.

3. RF EXCITATION
Flip to Transverse



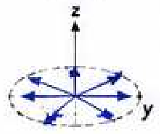
An RF pulse at the Larmor frequency tips M_0 away from z by flip angle α into the transverse plane (x-y).

4. TRANSVERSE MAGNETIZATION M_{xy}
Precessing Spins



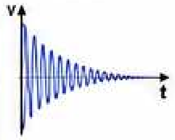
Spins precess in phase in the transverse plane, creating M_{xy} which can induce voltage in a receive coil.

5. DEPHASING
Spins Fall Out of Phase



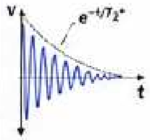
Slight spin–spin interactions (T_2) and field inhomogeneities (T_2*) cause spins to dephase.

6. FREE INDUCTION DECAY (FID)
Signal in Time Domain



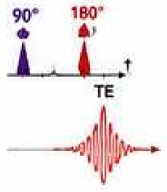
Immediately after the RF pulse, M_{xy} induces a decaying voltage (FID) in the receive coil.

7. T_2* DECAY
Includes Field Inhomogeneity



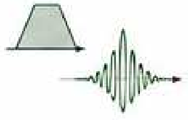
FID decays with T_2*, which is always shorter than or equal to T_2 due to local field variations (ΔB_0).

8. SPIN ECHO
180° Refocusing



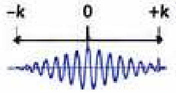
A 180° pulse reverses dephasing from static field inhomogeneity. Spins rephase and form an echo at TE. Decay reflects T_2.

9. GRADIENT ECHO (GRE)
Gradient Reversal



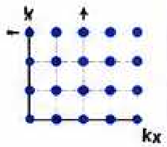
Gradient reversal causes rephasing and an echo at TE. GRE is sensitive to T_2* (no 180° pulse).

10. FREQUENCY – ENCODING
Spatial Information



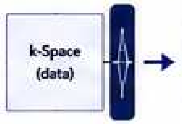
A frequency-encoding gradient maps position to frequency. The signal contains spatial information.

11. PHASE ENCODING
Fill k-Space



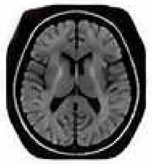
Phase-encoding gradient steps collect multiple lines of k-space.

12. FOURIER TRANSFORM
k-Space → Image



2D Fourier Transform converts k-space data into a spatial image.

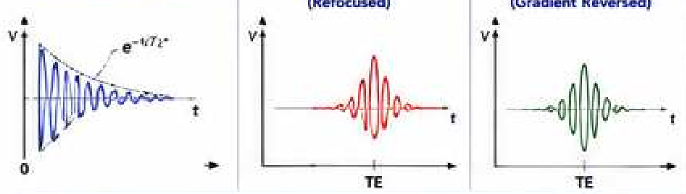
13. MR IMAGE
The Result



Contrast depends on sequence parameters and tissue properties (T_1, T_2, PD, T_2*).

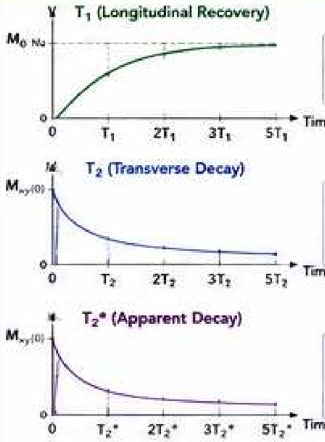
CREATION OF MAGNETIZATION → SIGNAL GENERATION (TIME DOMAIN) → ECHO FORMATION → SPATIAL ENCODING (K-SPACE) → IMAGE RECONSTRUCTION

TIME DOMAIN SIGNALS



	FID (After 90° Pulse)	SPIN ECHO (Refocused)	GRADIENT ECHO (GRE) (Gradient Reversed)
Rephasing Mechanism	None	180° RF pulse	Gradient reversal
Decay Constant	T_2* (includes ΔB_0)	T_2 (no ΔB_0)	T_2* (includes ΔB_0)
Echo Time (TE)	—	At TE	At TE
Signal Amplitude	Highest at t = 0	Higher (longer TE)	Lower (faster decay)
Sensitivity to Field Inhomogeneity	High	Low (static only removed)	High
Typical Use	Not used for imaging	SE, TSE, FSE	GRE, FLASH, SPGR

RELAXATION SUMMARY



M_z recovers toward M_0 with time constant T_1.

M_{xy} decays with true spin–spin relaxation (time constant T_2).

M_{xy} decays faster with T_2* due to field inhomogeneity.

Always: $T_2* \le T_2 \le 2T_1$

SEQUENCE COMPARISON AT A GLANCE

Feature	Spin Echo (SE)	Gradient Echo (GRE)
Refocusing Method	180° RF pulse	Gradient reversal
180° Pulse	Yes	No
Echo Type	Spin Echo	Gradient Echo
Decay Constant	T_2	T_2*
Sensitivity to ΔB_0 (Inhomogeneity)	Low	High
SNR (for same TE)	Higher	Lower
Imaging Speed	Slower	Faster
Tissue Contrast	T_2-weighted (via TE)	T_2*-weighted (via TE)
Common Sequences	SE, TSE, FSE	GRE, FLASH, SPGR, MPRAGE
Best For	High SNR, robust to inhomogeneity	Fast imaging, short TE, susceptibility contrast (e.g., SWI)

KEY TAKEAWAYS

- MRI begins with the hydrogen proton and its spin in B_0.
- RF excitation tips magnetization into the transverse plane.
- Precessing spins induce a voltage in the receive coil.
- Spins dephase due to T_2 (true) and T_2* (field inhomogeneity).
- FID is the raw signal right after excitation.
- 180° pulse refocuses static field inhomogeneity → Spin Echo.
- Gradient reversal refocuses spins → Gradient Echo.
- Echo amplitude decays: SE with T_2, GRE with T_2*.
- Spatial encoding with gradients fills k-space.
- Fourier Transform converts k-space data to an image.
- Contrast depends on sequence parameters and tissue properties (T_1, T_2, PD, T_2*).

ENGINEERING INSIGHT

- Shim quality (B_0 uniformity) directly affects T_2* and GRE.
- Gradient performance (linearity, timing) is critical for k-space accuracy and image quality.
- RF system (amplitude, phase, stability) determines flip angle accuracy and signal consistency.
- Coil performance (preamp, tuning, decoupling, Q) affects SNR and image uniformity.
- Environmental factors (temperature, vibrations) can impact stability and image quality.

FIELD SERVICE TIPS

- Verify shim before advanced sequences.
- Check gradient calibration and timing.
- Confirm RF output power and flip angle calibration.
- Inspect RF cables, preamps, and coil connections.
- Monitor for eddy current or vibration issues.
- Use phantom scans to baseline performance.
- Document and trend key QA metrics.

REAL-WORLD ANALOGY

Think of a group of runners starting together.

No 180° Pulse (Raw FID)

They spread out and never return.

Spin Echo (180° Pulse)

The 180° pulse makes them turn around and meet again at TE (echo).

Gradient Echo (GRE)

Changing the path (gradient) makes them meet at TE.

WHY IT MATTERS

- SE gives reliable images with high SNR.
- GRE enables fast imaging and specialty contrasts (susceptibility, flow, functional).
- Understanding this physics helps you troubleshoot, optimize, and maintain every MRI system.
- Good physics knowledge = better service, better uptime, better patient care.

QUICK REFERENCE

$\gamma/2\pi$ (proton) .... 42.576 MHz/T
Larmor Frequency .... $f_0 = \gamma/2\pi \cdot B_0$
T_1 = longitudinal relaxation time
T_2 = transverse relaxation time (spin–spin)
T_2* = apparent transverse relaxation time (includes ΔB_0)
TE = time to echo
k-space = spatial frequency domain
Image = 2D Fourier Transform of k-space data